

eAndon & TAKT Dashboard - User Acceptance Test (UAT)

Version: 6.0
Created: January 17, 2026
Author: Ivan Paudice
Status: In Execution - Intensive UAT Phase

Changelog from V5

- REMOVED:** EA-04 (Configure tiles per machine) - Feature hypothesis no longer needed
- ADDED:** EA-04a/b/c - Grey state transitions for planned stops
- ADDED:** EA-12a - CC field requirement in escalation emails
- ADDED:** EA-19a - Simultaneous failures to different support groups
- ADDED:** EA-26/27/28 - User management and permissions tests
- UPDATED:** EA-08 - Clarified "immediate" email timing expectation (~1 min acceptable)

Overview

This document defines the User Acceptance Test cases for the eAndon and TAKT dashboard system. All test cases must pass before any dashboard can be deployed to the shop floor.

Test Environment Requirements

- Access to Grafana dashboard configuration tool
- Test cell with configurable parameters (use shortened times for simulation)
- Email system access to verify notifications
- Multiple test users representing different support groups

Test Status Legend

Symbol	Meaning
□	Not tested
✓	Passed
✗	Failed
🕒	In progress / Partial

Part 1: eAndon Dashboard UAT

1.1 Dashboard Creation & Parameters

ID	Test Case	Expected Result	Status
EA-01	Create new dashboard with machine names	All machine names display correctly	●
EA-02	Configure machine positions in grid layout	Machines in correct grid positions (e.g., 9x3)	●
EA-03	Modify machine name after creation	Name change reflects immediately	●

1.2 Machine State Changes

ID	Test Case	Expected Result	Status
EA-05	Change machine state Green → Red	Visual indicator changes immediately to Red	●
EA-06	Change machine state Red → Green	Visual indicator changes immediately to Green	●
EA-07	Change machine state Green → Yellow	Visual indicator changes immediately to Yellow	●

EA-04a: Change machine state Green → Grey (NEW)

Field	Value
Expected Result	Visual indicator changes to Grey indicating a planned stop
Status	☐ Not tested

Context:

Grey state represents planned stops (lunch breaks, meetings, scheduled maintenance). Unlike Red/Yellow, Grey states should NOT trigger any escalation emails because no intervention is needed. This is the expected state during pause periods.

EA-04b: Change machine state Grey → Green (NEW)

Field	Value
Expected Result	Visual indicator returns to Green when planned stop ends
Status	☐ Not tested

Context:

When a planned pause ends, machines should automatically return to Green state. The transition should be smooth and the timer should resume from where it was frozen.

EA-04c: Verify Grey state does NOT trigger alerts (NEW)

Field	Value
Expected Result	No escalation emails are sent when a machine is in Grey state
Status	☐ Not tested

Context:

Critical test to ensure the system distinguishes between unplanned stops (Red/Yellow = send alerts) and planned stops (Grey = no alerts). If Grey state triggers emails, support groups will receive unnecessary notifications during every break.

1.3 Email Notification System

ID	Test Case	Expected Result	Status
EA-08	Immediate email on Red/Yellow state	Email sent within ~1 minute (0-min threshold)	
EA-09	Email group filtering - Maintenance	Red/Yellow for Maintenance → email ONLY to Maintenance	
EA-10	Email group filtering - Security	Red/Yellow for Security → email ONLY to Security	
EA-11	Email group filtering - Quality	Red/Yellow for Quality → email ONLY to Quality	☐
EA-12	Email group filtering - Other groups	Each problem type routes to designated group	☐
EA-13	First escalation email timing	After X minutes, escalation sent	
EA-14	Second escalation email timing	After Y minutes, second level escalation sent	☐
EA-15	Escalation chain completion	Full chain executes per configured times	☐

EA-12a: CC field in escalation emails (NEW - Feature Request)

Field	Value
Expected Result	Escalation emails include CC recipients for awareness/visibility
Status	☐ Not tested (Feature not yet implemented)

Context:

During the January 17 UAT session, Antonio identified a need for CC recipients on escalation emails. The use case is that supervisors or managers want to be aware of issues without being the primary responders. Currently, MotorAdmin only supports a single "To" address field (comma-separated for multiple recipients). A separate CC field would allow better email routing.

Action Required:

Francesco to evaluate adding a CC field to the escalation configuration in MotorAdmin.

1.4 eAndon Analytics & Data Verification

ID	Test Case	Expected Result	Status
EA-16	Create support request	Request logged with timestamp, type, machine ID	
EA-17	Acknowledge support request	Acknowledgment stops escalation timer	☐
EA-18	Close support request	Request closed, machine → Green, duration logged	
EA-19	Multiple simultaneous requests	System handles multiple Red states independently	☐
EA-20	Link from main dashboard to analytics	Click navigates to analytics dashboard	☐
EA-21	Pie chart displays correctly	Shows Red/Yellow/Green distribution for last 7 days	☐
EA-22	Drill-down on Red segment	Clicking Red shows table of all Red incidents	☐
EA-23	Drill-down on Yellow segment	Clicking Yellow shows table of all Yellow incidents	☐
EA-24	Incident detail table	Table shows: timestamp, machine, duration, problem type	☐
EA-25	Data accuracy verification	Analytics data matches actual logged incidents	

EA-19a: Simultaneous failures to different groups (NEW - Critical)

Field	Value
Expected Result	Two machines fail simultaneously with different problem types; each support group receives ONLY their relevant email
Status	☐ Not tested

Context:
This is the key test for validating email group filtering. The scenario: Machine A goes Red for Maintenance issue at the same moment Machine B goes Red for Security issue. Expected behavior: Maintenance group receives email about Machine A only; Security group receives email about Machine B only. If both groups receive both emails, the filtering bug is not resolved.

Technical Note (from January 17 meeting):
Francesco explained that the query logic creates emails only when each row matches both the department AND severity with corresponding addresses. Theoretically, simultaneous failures should not cause mix-ups, but this needs explicit verification.

1.5 User Management & Permissions (NEW Section)

EA-26: SSO login (NEW)

Field	Value
Expected Result	Users can authenticate using GE corporate SSO
Status	 Passed

Context:
SSO integration ensures that factory users can log in with their existing GE credentials without needing separate Grafana accounts. This is a compliance requirement for enterprise systems.

Execution Notes (January 17, 2026):
Antonio successfully logged into Grafana using his GE SSO credentials. The system recognized his user profile and applied the correct permissions based on his team membership.

EA-27: Team-based permissions (NEW)

Field	Value
Expected Result	Users see only the dashboards assigned to their team (e.g., Pomigliano F4)
Status	 In Progress

Context:
Factory floor users should have view-only access to their site's dashboards. This prevents accidental modifications while allowing operators to monitor production. Teams are configured per site (e.g., "Pomigliano", "Rivalta", "Brindisi").

Execution Notes (January 17, 2026):
Francesco demonstrated the team configuration in Grafana. Two teams were set up: "Administrators" (can edit all dashboards) and "Pomigliano" (view-only access to Pomigliano dashboards). Each team can have a default starting dashboard configured.

EA-28: Administrator access (NEW)

Field	Value
Expected Result	Administrators can edit dashboards; non-administrators have view-only access
Status	● Passed

Context:
Only designated administrators (IT, Digital Operations) should be able to modify dashboard configurations. Regular factory users must have read-only access to prevent accidental changes to production monitoring systems.

Execution Notes (January 17, 2026):
Verified the difference between admin and viewer roles. Administrators can modify dashboard configurations, change variables, and save changes. Regular users can only view the dashboard without modification capabilities.

Part 2: TAKT Dashboard UAT

2.1 Dashboard Creation & Configuration

ID	Test Case	Expected Result	Status
TK-01	Create new TAKT dashboard	Dashboard created with all required fields	☐
TK-02	Configure TAKT time (cycle time)	TAKT time parameter accepted and stored	☐
TK-03	Configure target pieces per shift	Target count displayed correctly	☐
TK-04	Configure shift schedules	Three shifts configured with correct times	☐

2.2 Visual Display

ID	Test Case	Expected Result	Status
TK-05	Timezone - Italy/Rome	All times display in Italy timezone (NOT UTC)	☐ ⚠ P0
TK-06	Dashboard title displays	Correct cell/line name shown	☐
TK-07	Current shift indicator	Correct shift (1st/2nd/3rd) displayed	☐
TK-08	Piece count display	Current piece count accurate and updating	☐
TK-09	Target piece count display	Target for current time shown correctly	☐
TK-10	Flow time display	Time per piece calculated and displayed	☐
TK-11	Progress bar display	Bar shows progress toward shift target	☐
TK-12	Progress bar color coding	Green (on target), Yellow (warning), Red (behind)	☐
TK-13	Variance indicator	Shows +/- pieces vs expected	☐
TK-14	Historical histogram	Shows variance distribution	☐

2.3 Shift Management

ID	Test Case	Expected Result	Status
TK-15	Shift 1→2 transition	Dashboard resets correctly at shift change	☐
TK-16	Shift 2→3 transition	Dashboard resets correctly at shift change	☐
TK-17	Shift 3→1 transition (day change)	Dashboard resets correctly at midnight crossing	☐ ⚠ P0
TK-18	Night shift handling (22:00-06:00)	Dashboard handles day boundary correctly	☐ ⚠ P0
TK-19	Shift totals persistence	Previous shift data retained for reporting	☐

2.4 Pause Management

ID	Test Case	Expected Result	Status
TK-20	Programmatic pause activation	Pause triggered at configured time	☐
TK-21	Tracking freeze during pause	TAKT counter pauses, variance frozen	☐
TK-22	Pause end and resume	Tracking resumes correctly after pause	☐
TK-23	Multiple pauses per shift	All configured pauses execute correctly	☐
TK-24	Pause time configuration	Pause times can be set per cell requirements	☐

2.5 Piece Tracking & Formulas

ID	Test Case	Expected Result	Status
TK-25	Piece advancement - on time	Completing piece on time shows Green	☐
TK-26	Piece advancement - early	Completing piece early shows accumulated buffer	☐
TK-27	Piece advancement - late	Late piece shows Red, variance updates correctly	☐
TK-28	Accumulated delay scenario	Multiple late pieces accumulate correctly in variance	☐
TK-29	Recovery scenario	Fast pieces can recover variance (within limits)	☐
TK-30	Threshold calculations	Red/Yellow/Green thresholds calculate correctly per config	☐

2.6 Edge Cases

ID	Test Case	Expected Result	Status
TK-31	Double entry (misclick)	System handles gracefully, no zero-duration piece	☐
TK-32	Overproduction prevention	After target reached, button disappears	☐
TK-33	Pause interrupts active piece	Timer freezes/resumes correctly	☐
TK-34	Piece logged during pause	Overlay prevents logging or handles appropriately	☐
TK-35	Pause spans shift boundary	Predictable behavior when pause crosses 22:00	☐
TK-36	Daylight Saving Time transition	System handles DST correctly	☐
TK-37	Network outage mid-shift	Data queues or clear indication of lost time	☐
TK-38	Brindisi multi-day piece	Piece spanning multiple shifts tracks correctly	☐

2.7 Shift Meeting Support

ID	Test Case	Expected Result	Status
TK-39	Shift 1 totals	Bottom-right shows Shift 1 summary	☐
TK-40	Shift 2 totals	Bottom-right shows Shift 2 summary	☐
TK-41	Shift 3 totals	Bottom-right shows Shift 3 summary	☐

2.8 TAKT Analytics

ID	Test Case	Expected Result	Status
TK-42	Link to analytics from TAKT	Navigation to analytics dashboard works	☐
TK-43	Analytics data accuracy	TAKT analytics match logged production data	☐

Part 3: Cross-Module Tests

ID	Test Case	Expected Result	Status
CM-01	eAndon Grey during TAKT pause	When pause active, eAndon Grey, TAKT frozen	☐
CM-02	End of pause synchronization	Both modules resume simultaneously	☐

Execution Log

Session 1: January 17, 2026 (09:00-10:15)

Participants: Francesco Salamone (Capgemini), Antonio Fusco (GE), Ivan Paudice (GE)

Summary:

- Created test line NGV5_1C with 8 machines
- Verified SSO login and team-based permissions
- Tested all state changes (Green/Red/Yellow) - all working
- Configured email escalation with multiple thresholds (0/1/2/3/4/5 min)
- Tested two departments (Maintenance, Security) with different severities
- Emails arriving successfully (both immediate and escalated)
- "Immediate" email took ~1 minute - acceptable

Issues Found:

1. CC field missing in escalation configuration (feature request)
2. Tooltips on hover - acceptable (touch screens on shop floor)

Next Session: January 17 (afternoon) or January 18 (15:00-16:00)

Test Summary

Section	Passed	In Progress	Not Tested	Total
eAndon	10	6	12	28
TAKT	0	0	41	41
Cross-Module	0	0	2	2
Total	10	6	55	71